User Behavior Mapping and AI-Powered Personalized Recommendations

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Table of Contents

List of Abbreviations

AI: Artificial Intelligence

BPR: Bayesian Personalized Ranking

CB: Content-Based Filtering

CF: Collaborative Filtering

CPU: Central Processing Unit

CRISP-DM: Cross-Industry Standard Process for Data Mining

CTR: Click-Through Rate

DCG: Discounted Cumulative Gain

DL: Deep Learning

GCN: Graph Convolutional Network

GNN: Graph Neural Network

GRU: Gated Recurrent Unit

H: Average Hamming Distance

I: Intra-Similarity

LGCNHS: Light Graph Convolutional Recommendation Algorithm Based on Hybrid Spreading

LLM: Large Language Model

LSTM: Long Short-Term Memory

ML: Machine Learning

NDCG: Normalized Discounted Cumulative Gain

NGCF: Neural Graph Collaborative Filtering

NLP: Natural Language Processing

ProbS: Probabilistic Spreading

HeatS: Heat Spreading

HybridS: Hybrid Spreading

RS: Recommendation System

RNN: Recurrent Neural Network

SBRS: Session-Based Recommendation System

SVD: Singular Value Decomposition

UBM: User Behavior Mapping

List of Tables

List of Figures

1. Introduction

With an ever-growing digital marketplace, online retailers continue to rely on complex algorithms to navigate the seemingly endless array of available items. As competition in this sector increases, it has become essential for retailers to provide every consumer with a customized experience to retain loyalty and maximize profits. However, in the wake of rising consumer expectations for relevance, existing recommendation models are experiencing significant limitations. The majority of online retailers struggle to present users with relevant product recommendations because they rely on static methods that depend heavily on historical transaction data (Batmaz et al., 2022). Consequently, consumer interaction remains limited, leading to numerous missed sales opportunities as these systems fail to capture the subtlety of a user's intent at any given moment or the fleeting nature of an active browsing session (Konstan & Terveen, 2021).

Modernizing the way e-commerce platforms understand their customers are Machine Learning (ML) systems and User Behavior Mapping. Advanced analytics can now interpret a complex web of implicit signals, such as dwell time, clickstream patterns, and mouse trajectories, to construct a vibrant, dynamic user profile (Klumppet et al., 2022). These AI-driven tools have the potential to evaluate huge volumes of behavioral data in real-time, discovering minute patterns and shifts in curiosity that static processes inevitably miss (Wei, He, & Gan, 2022). Furthermore, the integration of deep learning techniques enables systems to predict users' needs with unprecedented accuracy, transforming the passive catalog into a forward-thinking, intelligent sales agent that adapts to immediate context rather than just past purchases (Afchar & Cagnon, 2022).

Although significant progress has been made, essential challenges remain. While collaborative filtering is effective for established users, it struggles significantly with the 'cold start' problem for new visitors who lack a transaction history (Wu et al., 2023). Furthermore, current analysis often misses the full scope of customer behavior, failing to effectively integrate granular interaction forms, such as scrolling velocity or hesitation, to improve the accuracy of AI-based recommendations. It is, therefore, essential to determine how granular behavioral information can be properly synthesized with high-tech automated reasoning models to construct systems that are not only reactive but truly adaptive to the customer's current situation (Lin et al., 2022).

This thesis aims to answer the research question: 'How can AI-driven user behavior mapping improve the accuracy and relevance of personalized product recommendations in online retail platforms?' It investigates the synergies between computational analysis and machine learning, examining how mapping intricate user movements can bridge the gaps left by static recommendation models. The seminar component presents a comprehensive literature review, covering personalized recommendation frameworks and the crucial dimensions of user behavior mapping, while also detailing the machine learning algorithms used to model these complex behaviors. The thesis component then extends this foundation by implementing and evaluating the Light Graph Convolutional Recommendation Algorithm Based on Hybrid Spreading (LGCNHS), as proposed by Duan et al. (2025), on the MovieLens dataset.

Section 2 presents a comprehensive literature review organized into four domains: personalized recommendation systems, artificial intelligence techniques, the interrelationship between AI and user behavior mapping, and the current state of the art. Section 3 identifies the specific research gap. Section 4 presents the complete methodology, including dataset description, preprocessing, model architecture, training configuration, and evaluation design. Section 5 presents and discusses the experimental results. Section 6 concludes the thesis and outlines directions for future research.

2. Literature Review

The literature surrounding e-commerce personalization has evolved significantly over the past decade, shifting from broad demographic segmentation to individual behavioral targeting. Early research primarily focused on static user attributes, but the exponential growth of digital touchpoints has necessitated a transition toward dynamic, real-time data analysis (Konstan & Terveen, 2021). Contemporary studies now emphasize the critical role of 'implicit feedback', such as dwell time and navigation paths, as a richer source of user intent than explicit ratings alone (Wei, He, & Gan, 2022). This paradigm shift has established a new foundation for online retail strategies, where the ability to interpret vast streams of unstructured behavioral data is the primary determinant of competitive advantage and customer retention (Batmaz et al., 2022).

This chapter critically examines the technological and theoretical advancements that enable this shift, categorizing the existing body of knowledge into four distinct domains. It begins by analyzing the evolution of personalized recommendation systems, contrasting traditional collaborative filtering with modern deep learning architectures (Batmaz et al., 2022). Subsequently, the review explores the specific mechanisms of User Behavior Mapping (UBM), detailing how granular actions are harvested and processed to predict purchase probability (Klumppet et al., 2022). Finally, the chapter synthesizes these fields to present the current 'State of the Art,' highlighting emerging trends such as Large Language Model integration and identifying the persistent gaps in balancing computational efficiency with recommendation accuracy (Gao et al., 2023).

2.1 Personalized Recommendation Systems and User Behavior Mapping in E-Commerce

The emergence of e-commerce has been fundamentally tied to the exponential growth in capabilities to process and filter large volumes of data, ensuring consumers are directed to appropriate products and services. The ability to generate sales from a user's passive view of an online store remains a significant challenge faced by today's retailers; therefore, personalization through Recommendation Systems (RS) has become increasingly important as a primary driver of revenue and customer engagement (Batmaz et al., 2022). This section outlines the primary forms of recommendation systems, which have evolved significantly with the integration of Deep Learning techniques. It further describes the shortcomings of current methods, specifically regarding the dynamic nature of user preferences, and discusses User

Behavior Mapping (UBM) as a granular method that addresses these gaps by providing predictive insight into consumer intentions (Klumpp et al., 2022).

2.1.1 Overview of Recommendation Systems in Online Retail

Deep learning methods based on the primary values of Collaborative Filtering (CF) and Content-Based Filtering (CB) dominate current recommendation structures (Batmaz et al., 2019). Collaborative filtering generates recommendations based on the exchanges between users and items, while content-based filtering concentrates on the properties of items to propose merchandise (Ricci, Rokach & Shapira, 2022). While these authoritative approaches are distinct, the current state-of-the-art includes hybrid models using neural networks to process and integrate diverse statistics for superior performance. For instance, the large e-commerce giant Alibaba employs a deep learning model that integrates consumer behavior sequences with rich item characteristics to generate highly accurate, real-time recommendations (Zhou et al., 2021).

However, even such advanced frameworks face persistent 'cold start' problems, whereby performance suffers for recent users or items with no interaction history. Recent exploration attempts to decipher this problem by employing tactics like Graph Neural Networks (GNNs) to model the bonds among items (Wu et al., 2022). Beyond technical metrics, the rise of deep learning has brought new challenges regarding clarity; the intricate, 'black-box' environment of many neural models makes it difficult to perceive why a specific recommendation is made. This lack of transparency can significantly diminish user confidence, as customers are less likely to engage with suggestions they do not understand (Zhang & Chen, 2021).

Furthermore, these systems create and reinforce a 'filter bubble' by continuously proposing items similar to a customer's previous interactions, thus limiting their exposure to new content. Deldjoo et al.'s substantial analysis (2021) points out that the recommendation process may accidentally produce a feedback loop which magnifies the current bias in the statistics, leading to unjustified exposure for certain product or consumer groups. For instance, if a model is trained on historical data where specific demographics prefer technology products, it can create prejudiced experiences by enforcing those stereotypes. Consequently, the main focus of current exploration is addressing these ethical aspects, emphasizing the development of explainable and fair methods.

2.1.2 Limitations of Traditional Recommendation Methods

A significant disadvantage of traditional recommendation models is their inability to capture the active, short-term intent of users during a single browsing session. These systems frequently rely on a user's full historical interaction log, which often fails to reflect their specific, current shopping mission (Afchar & Cagnon, 2022). Consequently, Session-based Recommendation Systems (SBRS) have emerged as a pivotal research area, focusing exclusively on sequential events within the current session to predict subsequent actions adapted to changing contexts. For instance, a consumer browsing for a child's gift displays a distinct intent compared to their usual behavior, a nuance that session-based models capture effectively by evaluating the immediate clickstream (Wang et al., 2021).

While session-based models address immediate context, scholars argue that completely ignoring long-term user history is a lost opportunity. Recent hybrid models seek stability by applying attention mechanisms to weigh a consumer's sustained profile against their immediate actions, thereby providing more comprehensive recommendations (Li et al., 2021). However, beyond user intent, e-commerce faces the persistent 'item cold start' obstacle. Traditional models based on interaction records cannot effectively suggest novel items until they have been seen and rated by a sufficient number of users (Wu et al., 2022). This limitation is particularly acute in fast-moving industries, such as electronics, where new inventory must be promoted immediately.

To mitigate the cold start issue, state-of-the-art approaches increasingly employ Graph Neural Networks (GNNs). These models learn the embeddings of new items based on attributes such as brand and category, associating them with existing objects to enable immediate recommendation (Shi et al., 2022). Despite this technical progress, critical studies suggest the practical effectiveness of these complex models may be overstated. Research by Dacrema, Boglio, and Jannach (2021) demonstrates that simpler, well-tuned baseline methods can often outperform complex deep learning models, indicating that increased complexity does not always guarantee superior performance in real-world production environments.

2.1.3 Key Dimensions of User Behavior Mapping

To overcome the restrictions of sparse explicit feedback, such as star ratings, state-of-the-art frameworks rely on User Behavior Mapping (UBM) to analyze a rich stream of implicit signals. This process involves tracking and modeling the digital footprint of a consumer, including clicks, search queries, dwell time, and mouse movements, to construct an active, dynamic profile (Klumppet et al., 2022). The user behavior sequence, or clickstream, is particularly powerful for identifying immediate intent. For instance, a buyer who views a product, examines its details, and then returns to a search page demonstrates a clear pattern of comparison and high purchasing probability, which can be effectively captured by sequential recommendation models (Wang et al., 2021).

The primary value of these actions lies in the provision of a dense and continuous flow of information regarding client preferences. Unlike explicit evaluations, which are rare, implicit responses are generated with every single customer interaction, allowing for a more nuanced understanding of engagement (Koren & Rendle, 2021). A critical metric in this domain is 'dwell time,' or the duration spent on a page. A recent study by Wei, He, and Gan (2022) integrated dwell time into a neural network architecture and found it substantially improves the accuracy of purchase predictions. By treating time as a proxy for interest, models can differentiate between accidental clicks and genuine consideration, enhancing the relevance of suggestions.

However, scholars caution against the inherent ambiguity of these implicit signals. Konstan and Terveen (2021) warn that behavioral cues can be easily misinterpreted; a long dwell time may indicate genuine fascination, but it could just as easily signal user confusion or an off-screen distraction. This ambiguity necessitates advanced models that interpret signals contextually rather than as definitive isolated facts. Furthermore, the accumulation of such granular behavioral data raises significant ethical dilemmas regarding privacy. This has directed research toward privacy-preserving approaches, such as Federated Learning, which aims to train models without centralizing sensitive client information (Yang et al., 2021).

2.1.3.1 Behavioral Indicators and Their Predictive Value

Clickstream information and shopping cart events are among the most powerful behavioral indicators available to online retailers. Assessment of the sequence of user clicks during a session is a cornerstone of state-of-the-art session-based recommendations. Transformer models, originally optimized for Natural Language Processing (NLP), have demonstrated significant success in understanding the 'grammar' of a customer's shopping journey in order to predict their subsequent click (Sun et al., 2021). By treating a browsing session as a sentence and products as words, these advanced models capture complex sequential dependencies, allowing the system to anticipate a user's next move with high precision. This granular analysis moves beyond static preferences, enabling the platform to adapt to the specific context of the current visit.

The abandonment of a shopping cart is another critical event in the lower buying funnel. Recent machine learning models can anticipate cart abandonment with high accuracy by studying real-time behaviors, allowing for timely interventions such as the provision of a discount or a customer assistance chat (Makdissi & El-Hajj, 2020). For instance, analysis might reveal that users who hesitate at shipping costs are prime candidates for a free shipping offer. However, researchers argue against treating every abandonment as a lost sale. A study by Valaskova, Kliestik, and Gajdosik (2021) suggests that many users utilize shopping carts merely as wish lists or comparison tools; consequently, aggressive retargeting campaigns based on this signal could be perceived as intrusive and negatively affect the long-term relationship between the buyer and the merchant.

2.1.3.2 Integrating Multi-Channel User Data for Holistic Insights

To build a truly inclusive customer profile, retailers must adopt an omnichannel approach that integrates data across all touchpoints, including the web, mobile applications, social media, and physical stores. This methodology is crucial for providing a uniform and personalized buyer experience within a fragmented retail environment (Gao et al., 2021). The primary objective is to create a unified view of the customer that enables seamless personalization; for instance, a consumer who browses a product on their mobile device should encounter relevant recommendations on their desktop later that day. Recent analysis demonstrates that effective omnichannel integration significantly enhances consumer brand equity and loyalty, as it creates a cohesive narrative around the user's journey rather than disjointed interactions (Kumar, Anand, & Song, 2021).

Despite the clear benefits, the implementation of such systems faces significant technical and organizational hurdles, primarily regarding the breaking of data silos. A study by Rodrigues, D'Souza, and Faria (2022) highlights that many institutions struggle with the complex data integration and real-time analytics capabilities required for a true omnichannel strategy. Furthermore, this level of cross-platform data collection escalates privacy concerns, necessitating robust data governance frameworks. To maintain consumer confidence in an increasingly data-driven world, retailers must prioritize user consent and transparency, ensuring that the aggregation of personal data across channels complies with evolving privacy standards (Mansurali et al., 2024).

2.2 Artificial Intelligence

Artificial Intelligence (AI), specifically through Machine Learning (ML) and Deep Learning (DL), provides the computational engine required to execute the complex user behavior mapping and personalization discussed in the previous section. These techniques enable platforms to transcend static, rule-based systems by learning from massive datasets to identify intricate patterns in user behavior and generate predictive decisions in real-time (Batmaz et al., 2022). Unlike traditional methods, these dynamic systems continuously adapt to new data, ensuring that the recommendation logic evolves alongside changing consumer trends.

In this context, the primary function of AI is to transform raw behavioral data, such as clicks, dwell time, and purchase history, into actionable insights that drive precise recommendations (Klumpp et al., 2022). For example, an AI model can analyze a user's navigation path to predict purchase probability and dynamically adjust the interface to maximize conversion rates (Wei et al., 2022). However, the implementation of these sophisticated models faces challenges; researchers note that complex deep learning architectures do not always outperform simple, well-tuned baseline algorithms in real-world scenarios, raising questions about the trade-off between computational complexity and tangible value (Dacrema et al., 2021).

2.2.1 Machine Learning Algorithms in Recommendation Systems

Machine learning algorithms, broadly classified into supervised and unsupervised categories, form the core of data-driven personalization. In supervised learning, algorithms are trained on labeled data to predict specific outcomes through techniques like classification and regression (Batmaz et al., 2022). Classification assigns categorical labels, such as 'like' or 'dislike,' while regression models predict continuous numeric values, similar to the detailed star ratings a user might assign to an item (Al-Ghuribi & Noah, 2021). A practical application of this is predicting shopping cart abandonment by analyzing session features, which allows retailers to offer timely support or incentives (Makdissi & El-Hajj, 2020). However, reliance on explicit feedback like ratings is increasingly viewed as a limitation due to data sparsity (Koren & Rendle, 2021).

Unsupervised learning, particularly clustering, is utilized to detect hidden structures within datasets that lack predefined labels. Methods such as K-Means are frequently employed in e-commerce to group users into distinct segments based on shared characteristics like buying habits, browsing demeanor, or demographics (Saha et al., 2021). This approach enables segment-level personalization, where recommendations are adapted to a persona, such as 'discount seekers' or 'luxury shoppers', rather than an individual user (Klumpp et al., 2022). For example, a fashion retailer might use clustering to identify groups preferring specific styles, subsequently using that insight to target relevant products across broader marketing campaigns (Ricci et al., 2022).

Despite their utility, traditional machine learning methods face significant limitations regarding flexibility. A major drawback of conventional clustering is its rigidity; a consumer is typically assigned to a single exclusive group, which fails to capture the reality that user interests are often multifaceted and span across different categories. This lack of fluidity means the system cannot easily adapt when a user's intent shifts during a specific session. This dilemma, combined with the difficulty of modeling complex, non-linear interactions using standard algorithms, has driven the industry toward more advanced Deep Learning approaches that can process raw implicit signals for dynamic profiling (Batmaz et al., 2022).

2.2.1.1 Supervised vs. Unsupervised Learning Applications

Supervised learning approaches are fundamentally predictive and goal-oriented, as they acquire a mapping function from labeled input statistics toward a known final outcome. These models are ideal for tasks with clear, measurable goals, such as predicting purchase intent or click-through rates (Wei et al., 2022). For instance, regression models project continuous variables like explicit user ratings (e.g., 1 to 5 stars), which are then used to rank items for recommendation (Al-Ghuribi & Noah, 2021). Additionally, classification models predict binary outcomes, such as identifying users with a high liability of churning or abandoning their shopping carts, thereby enabling timely retail intervention (Makdissi & El-Hajj, 2020). However, the primary constraint of supervised learning is its dependency on large volumes of high-quality, manually labeled data, which is often expensive and difficult to obtain in real-world e-commerce conditions (Koren & Rendle, 2021).

In contrast, the objective of unsupervised learning is exploratory, aiming to reveal hidden patterns and structures within unlabeled datasets. A primary application is customer segmentation, where clustering algorithms establish distinct user groups based on their behavioral actions rather than predefined categories (Saha et al., 2021). This approach grants platforms the ability to penetrate their customer base and discover organic growth patterns free from the bias of pre-existing labels (Klumppet et al., 2022). Beyond user grouping, unsupervised techniques like graph-based methods are employed to identify latent relationships between items in a catalog. These structural insights are then utilized to influence the recommendation of similar or complementary products without requiring explicit training data for every pair (Wu et al., 2022).

Despite these advantages, interpretation and appraisal remain significant obstacles to the effective deployment of unsupervised learning. While a model may successfully recognize a distinct statistical group, it cannot inherently explain what that group represents, often requiring human analysts to study the output to derive semantic meaning. Furthermore, traditional clustering methods often treat user interests as static properties, which becomes problematic in dynamic retail environments. As a result, these models may struggle to adapt to the highly active, session-based desires of the consumer, often failing to capture the fleeting intent that characterizes modern e-commerce browsing (Afchar & Cagnon, 2022). This limitation highlights the necessity for more advanced architectures capable of handling real-time variability.

2.2.1.2 Role of ML in Dynamic User Segmentation

Machine learning is essential for transitioning customer segmentation from static to dynamic systems. Traditional segmentation typically relies on static demographic statistics; however, modern ML models can segment users in real-time based on their immediate behaviors, including clickstreams, search queries, and interaction patterns (Klumpp et al., 2022). This capability enables a more flexible understanding of the customer, recognizing that their objectives may change drastically from one session to the next. For instance, a consumer usually classified as a 'discount shopper' based on purchase history might behave as a 'high-value gifter' during a holiday season. A dynamic ML model can identify this temporary shift in intent and adjust recommendations accordingly, ensuring relevance regardless of historical labels (Wang et al., 2021).

The primary advantage of dynamic segmentation lies in its direct connection to real-time personalization. By continuously re-evaluating a user's segment based on their most recent behavior, an ML-driven system ensures that recommendations remain aligned with the immediate context (Afchar & Cagnon, 2022). This approach is particularly effective in session-based recommendations, where the customer's 'segment' acts as a fluid representation of their current shopping mission derived from live in-session demeanor (Wang et al., 2021). For example, a model integrating dwell time can dynamically increase its confidence in a customer's interest for a specific product category, updating their segmentation profile and refining subsequent recommendations within the very same session (Wei et al., 2022).

However, the implementation of such high-fidelity tracking introduces significant challenges regarding privacy and infrastructure. The granular level of real-time monitoring required for instant re-categorization can be perceived as highly intrusive, raising ethical concerns that necessitate the development of privacy-preserving methods like Federated Learning (Yang, Fan, & Chen, 2021). Furthermore, this real-time capability imposes a heavy computational burden. Continuously running dynamic segmentation models for millions of concurrent active users requires a robust, low-latency infrastructure. This scalability requirement presents a major technological obstacle, often forcing a trade-off between the complexity of the segmentation model and the speed of the user experience (Batmaz et al., 2022).

2.2.2 Deep Learning Approaches for Modeling Complex Consumer Behaviors

While traditional machine learning algorithms are efficient, they often struggle to model the intricate, non-linear, and sequential nature of complex customer behavior. This is where Deep Learning (DL) approaches, utilizing multi-layer neural networks, demonstrate superior performance by automatically learning feature representations from raw data (Batmaz et al., 2022). For instance, Transformer models like BERT4Rec borrow architectures from Natural Language Processing (NLP) to treat a user's interaction history as a sentence, allowing them to predict the next 'phrase', or item, with extraordinary accuracy (Sun et al., 2021). However, the 'black box' nature of these powerful models remains a primary criticism; their internal decision-making processes are frequently opaque, making it difficult to debug errors, explain recommendations to users, or verify the framework for potential bias (Zhang & Chen, 2021).

Distinct Deep Learning architectures are optimized for different types of behavioral data. Recurrent Neural Networks (RNNs) and their variants, such as Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs), are intrinsically sequential, making them perfectly suited to model clickstream data and power session-based recommendations (Afchar & Cagnon, 2022). Furthermore, Graph Neural Networks (GNNs) have recently achieved high prominence due to their capability to model complex graphs of relationships between users, items, and attributes (Wu et al., 2022). For example, to generate embeddings for new products, a GNN can learn from the surrounding user-item interaction graph, successfully mitigating the persistent 'item cold-start' problem by leveraging attribute connections rather than historical interaction data (Shi et al., 2022).

Despite the theoretical advantages and increased precision reported in numerous studies, significant questions regarding practical viability call for caution. A critical, extensive study by Dacrema, Boglio, and Jannach (2021) demonstrated that many novel, complex Deep Learning models frequently fail to outperform simpler, well-tuned conventional methods when evaluated fairly. This finding highlights a significant gap between research claims and reproducible, real-world value, suggesting that complexity does not always equate to better performance. Consequently, the current research direction emphasizes not only the development of sophisticated architectures but also the rigorous benchmarking of these models against standard baselines to ensure they provide genuine operational improvements in e-commerce environments.

2.2.3 Context-Aware and Real-Time Adaptation Through AI-Driven Personalization

The evolution toward context-aware systems represents a critical frontier for AI-powered personalization. These systems adjust recommendations based not only on the customer's static profile but also on relevant situational circumstances, such as geographical location, time of day, or the specific device being used (Ricci et al., 2022). This adaptability is a fundamental component of a successful omnichannel strategy, designed to integrate data from all touchpoints to provide a seamless and contextually relevant experience (Gao et al., 2021). For instance, a client browsing on a mobile device during a morning commute requires a distinctly different recommendation strategy compared to the same client browsing on a desktop computer in the evening (Wang et al., 2021).

Despite the potential of context-aware personalization, data integration remains a primary impediment to its effective implementation. Many organizations struggle with the significant structural and technical constraints required to provide AI models with a comprehensive, real-time overview of the user's background (Büschken & Manz, 2023). Legacy systems often fragment data across isolated silos, separating mobile app logs from web transaction history, which prevents the algorithm from accessing the holistic context needed for accurate predictions. Consequently, without a unified data architecture, the AI cannot distinguish between a user's immediate intent and their long-term preferences, limiting the system's ability to deliver truly adaptive and relevant experiences in dynamic retail environments.

The ultimate goal of this technology is real-time adaptation, where the AI model updates its recommendations dynamically during a buyer's session. This is the defining attribute of Session-based Recommendation Systems (SBRS), designed to remain highly responsive to the buyer's evolving intent (Afchar & Cagnon, 2022). Advanced models utilize attention mechanisms to dynamically weigh a client's past choices against their most recent clicks, allowing the system to pivot its strategy on the fly (Li et al., 2021). For example, Alibaba employs deep learning to update ad and product recommendations within seconds of a new interaction (Zhou et al., 2021). However, critics argue that frameworks reacting too aggressively to individual, potentially accidental clicks can produce a jarring user experience, emphasizing the need for human-centered design to maintain trust (Konstan & Terveen, 2021).

2.3 Interrelationship Between AI, User Behavior Mapping, and Personalized Recommendations

While personalized recommendation systems (Section 2.1) and Artificial Intelligence (Section 2.2) are distinct domains, they are profoundly symbiotic in practice. User behavior mapping provides the high-volume, high-velocity data that serves as the essential 'fuel' for AI models (Klumpp et al., 2022). Conversely, AI acts as the 'engine' capable of processing this massive, often noisy behavioral data to identify meaningful patterns and generate accurate predictions (Batmaz et al., 2022). This synergy is the foundation of session-based recommendations, where sequential AI models analyze clickstream data to predict immediate intent (Afchar & Cagnon, 2022). However, this relationship is not without friction; the ultimate quality of an AI prediction is wholly dependent on the accuracy of the mapped behavioral data, which can often be ambiguous and difficult to interpret correctly (Konstan & Terveen, 2021).

2.3.1 How Behavior Mapping Enhances AI-Powered Recommendation Accuracy

User behavior mapping directly enhances the accuracy of AI-powered recommendations by addressing the persistent challenge of data sparsity. Traditional AI models were often limited to explicit feedback mechanisms, such as star ratings, which users provide infrequently, resulting in sparse and unreliable profiles (Batmaz et al., 2022). In contrast, behavior mapping enriches these profiles by capturing a continuous stream of implicit feedback, including clicks, page views, scrolling velocity, and mouse movements, thereby providing the AI with a significantly denser dataset for training and inference (Klumpp et al., 2022). This shift allows the system to generate insights based on actual engagement rather than waiting for sporadic user input.

The integration of specific behavioral indicators, such as dwell time, demonstrates this potential for improved precision. A study by Wei, He, and Gan (2022) revealed that incorporating dwell time into deep learning architectures significantly enhanced the model's ability to predict purchase intent compared to baseline methods. However, researchers caution against the inherent ambiguity of such implicit data; Konstan and Terveen (2021) argue that these signals are often 'noisy' and open to misinterpretation. For example, a long duration spent on a page could indicate high user interest, but it could equally signify confusion or an off-screen distraction, potentially leading the AI to draw incorrect conclusions about user preferences.

Furthermore, mapping sequential behavioral patterns enables AI models to transcend static user profiles and interpret dynamic user intent. By analyzing the specific order of actions within a clickstream, AI can infer a user's current shopping mission, distinguishing between phases such as 'exploratory research' and 'ready to purchase' (Afchar & Cagnon, 2022). This sequential data serves as the primary input for advanced deep learning architectures, such as Recurrent Neural Networks (RNNs) and Transformers, which process a browsing session similarly to a sentence in natural language (Wang et al., 2021). While models like BERT4Rec leverage this to predict the next logical interaction (Sun et al., 2021), critics note that real-world browsing is often chaotic and non-linear, making it difficult to extract a single, coherent intent from multi-tabbed, multi-goal sessions (Klumpp et al., 2022).

2.3.2 Synergies in Creating Adaptive, Context-Aware, and Dynamic E-Commerce Personalization

The true synergy between AI and behavior mapping is most evident in the creation of adaptive and context-aware personalization. This synergy is the technological foundation of a modern omnichannel strategy, where AI models integrate behavioral data from all customer touchpoints — including the website, mobile app, and even in-store interactions — to build a single, unified customer profile (Gao et al., 2021). This holistic, behavior-mapped profile allows the AI to deliver a consistent and adaptive experience; for example, an item abandoned in a shopping cart on a mobile app can trigger a personalized incentive on the user's desktop browser (Kumar et al., 2021). This seamless integration has been shown to enhance brand loyalty and customer equity (Gao et al., 2021). The primary barrier to this synergy, however, is often not the AI's capability but the organizational data silos that prevent the model from accessing a truly unified stream of behavioral data in real-time (Büschken & Manz, 2023).

2.3.2.1 Integration Models Between Behavioral and Algorithmic Systems

The most common integration model involves using mapped behavioral data as engineered features for traditional machine learning algorithms. In this approach, raw behavioral signals (e.g., number of clicks, average dwell time, items in cart) are aggregated and pre-processed to create a static feature vector for a user or session (Klumpp et al., 2022). This feature vector is then fed into a supervised learning algorithm, such as a classification model, to predict a specific outcome (Al-Ghuribi & Noah, 2021). A clear example is a cart abandonment prediction model, which uses features from a user's current session to classify them as 'likely to abandon' or 'likely to purchase,' allowing the system to intervene (Makdissi

& El-Hajj, 2020). The main limitation of this model is that the process of feature engineering is labor-intensive and often 'flattens' sequential data, losing the valuable context embedded in the order of the user's actions (Afchar & Cagnon, 2022).

To overcome this, more advanced integration models use deep learning to process raw behavioral sequences end-to-end, eliminating the need for manual feature engineering (Batmaz et al., 2022). In this model, the AI system (e.g., an RNN or Transformer) is fed the raw clickstream (e.g., [item_A, item_B, item_C]) directly, learning its own representations of the data (Wang et al., 2021). This allows the model to capture complex, non-linear patterns and dependencies within the user's behavior, as seen in models like BERT4Rec (Sun et al., 2021). A significant critique of these powerful end-to-end models is their 'black-box' nature. Because the model learns its own features, its internal logic becomes opaque, making it difficult to explain why a specific recommendation was made, which can in turn erode user trust (Zhang & Chen, 2021).

2.3.2.2 Emerging Trends: Generative AI and Predictive Analytics

A major emerging trend is the use of predictive analytics to move beyond a single 'next-click' recommendation and instead model a user's entire future journey. This involves using rich behavioral data to create a 'digital twin' of the consumer to simulate how they might react to different stimuli (Klumppet et al., 2022). Graph Neural Networks (GNNs) are at the forefront of this trend, as they integrate the behavior of all users to map the complex 'ecosystem' of user-item interactions (Wu et al., 2022). A GNN can use this map to predict how a user's interest will evolve, for example, suggesting a product from a different category that is a logical 'next step' in their journey (Shi et al., 2022). The challenge for these large-scale models, however, is their potential to create 'rich-get-richer' feedback loops, where the AI's predictions inadvertently make popular items even more popular, reducing diversity and fairness (Deldjoo et al., 2021).

The most disruptive trend is the integration of Generative AI, particularly Large Language Models (LLMs), into recommendation systems. This new paradigm shifts the focus from matching behavioral patterns to understanding user intent through natural language and reasoning (Tao et al., 2023). Instead of only analyzing what a user clicked, a GenAI system can understand why by processing product reviews, descriptions, and complex, conversational user queries. For example, a user could ask, 'Find me a warm, waterproof jacket that is also lightweight for hiking,' and the LLM could synthesize these multiple, complex attributes to find the ideal product (Sanner, 2024). This approach also holds promise for solving the 'black box' problem, as the LLM can generate a natural language sentence explaining why it recommended an item (Zhang & Chen, 2021). However, critics argue this technology is in its infancy and that LLMs are prone to 'hallucinating' facts and can perpetuate the biases present in their vast text training data (Tao et al., 2023).

2.4 State of the Art

The most significant recent advancement is the integration of Generative AI into recommendation pipelines. Gao et al. (2023) introduced 'Chat-REC,' a framework that bridges LLMs with traditional recommendation systems. By converting user profiles and historical interactions into natural language prompts, the system allows the LLM to infer user intent with contextual reasoning that statistical models lack. Similarly, Bao et al. (2023) proposed 'TALLRec,' which fine-tunes LLaMA-based models specifically for recommendation tasks using instruction tuning. These approaches demonstrate that LLMs can effectively function as recommendation engines even with limited training data (few-shot learning), although they suffer from high latency and 'hallucination' issues where non-existent items are recommended.

To better map complex user behavior beyond simple clicks, researchers are leveraging Graph Neural Networks (GNNs). Lin et al. (2022) developed a 'Multi-behavior Hypergraph Enhanced Transformer' (MBHT) which models multiple types of user actions (view, add-to-cart, purchase) as a hypergraph. This allows the system to capture high-order dependencies between different behaviors, significantly improving accuracy over single-behavior models. In a similar vein, Xia et al. (2023) proposed an automated self-supervised learning framework for graph recommendations. Their method automatically generates data augmentations to train the graph model, effectively solving the issue of noise in implicit feedback data, though the training process remains computationally expensive.

Recent research has also addressed the challenges of multi-domain learning and secure data sharing. Yang et al. (2024) introduced 'MLoRA,' a Multi-Domain Low-Rank Adaptive Network that adapts the popular 'LoRA' fine-tuning technique from LLMs to Click-Through Rate (CTR) prediction. By training domain-specific low-rank adapters while sharing a backbone model, they achieved state-of-the-art performance in scenarios where user data is fragmented across different e-commerce domains. Simultaneously, addressing the privacy concerns of data sharing, Hou and Zhang (2023) proposed 'BFRecSys,' a blockchain-based federated matrix factorization framework that decentralizes the aggregation of model updates, eliminating the single point of failure inherent in traditional federated learning servers.

Addressing the limitation of sparse user data, Contrastive Learning (CL) has emerged as a dominant methodology. Wu et al. (2023) introduced a Graph Contrastive Learning method that augments user-item interaction graphs by randomly dropping edges or nodes to create 'views' of the data. By forcing the model to recognize that these different views represent the same user, the system learns robust representations without needing massive amounts of labeled data. Wang et al. (2022) applied contrastive learning specifically to sequential recommendation, improving the model's ability to predict the next click in short, anonymous sessions. While highly effective for cold-start scenarios, these methods often struggle with 'false negatives' during the contrastive training phase.

Recent work has also focused on removing the bias inherent in historical behavior mapping. Zhang et al. (2023) proposed a causal inference framework to disentangle a user's true interest from conformity bias (following trends). By modeling the causal graph of user interactions, they improved the diversity of recommendations. Additionally, Li et al. (2023) explored prompt learning to guide models away from biased pathways. Table 1 below summarizes the key approaches, results, advantages, and disadvantages of the most relevant state-of-the-art works reviewed in this chapter.

Table 1: Summary of State-of-the-Art Approaches

Reference	Methodology	Result	Advantages	Disadvantages
Gao et al., 2023	Converts user history into prompts for ChatGPT/LLMs.	High zero-shot performance and conversational interactivity.	Contextual reasoning; highly explainable outputs.	High inference latency; prone to hallucinations.
Bao et al., 2023	Fine-tunes LLMs (LLaMA) for recommendation tasks.	Outperformed traditional CF in few-shot scenarios.	Effective generalization; requires very little training data.	Computationally expensive to fine-tune.
Yang et al., 2024	Multi-Domain Low-Rank Adaptive Network.	Superior CTR prediction across diverse domains.	Efficient parameter adaptation; solves negative transfer.	Requires careful tuning of rank parameters.
Hou & Zhang, 2023	Blockchain-based Federated Matrix Factorization.	Secure model updates without a central server.	Decentralized trust; tamper-proof model aggregation.	High latency due to blockchain consensus; computational overhead.
Lin et al., 2022	Models clicks/carts as hypergraph.	Significant improvement in conversion prediction.	Captures complex dependencies; utilizes implicit data.	Hypergraph construction is complex.

Xia et al., 2023	Automated Self-Supervised Learning.	SOTA accuracy on noise-heavy datasets.	Robust against noisy data; automates augmentation.	Training process is unstable.
Wu et al., 2023	Augmented views for self-supervision.	Superior performance in cold-start scenarios.	Solves data sparsity effectively; no labels required.	Risk of 'false negatives'.
Wang et al., 2022	Data augmentation for sessions.	Improved next-item prediction for short sessions.	Maximizes utility of short data; better sequence representations.	Risk of destroying sequence logic.
Zhang et al., 2023	Causal graphs to separate popularity bias.	Increased diversity and novelty.	Removes popularity bias; better generalization.	Strict causal assumptions required.
Li et al., 2023	Adapts pre-trained models via prompt learning.	Unified performance across multiple tasks.	Parameter efficient; versatile.	Sensitive to prompt design.

3. Research Gap

The increasing demand for hyper-personalized e-commerce necessitates adaptive engines, yet significant gaps regarding latency and scalability persist. While recent literature prioritizes Large Language Models and Graph Neural Networks for their reasoning capabilities, these architectures often struggle with high inference costs and hallucination risks in real-time environments (Gao et al., 2023; Bao et al., 2023). Furthermore, existing hypergraph models frequently overlook granular implicit signals like dwell time due to the prohibitive computational cost of constructing complex user graphs (Lin et al., 2022). Consequently, current methods fail to effectively balance deep semantic understanding with the speed required for online retail.

A further gap exists in the integration of physics-inspired spreading-based recommendation algorithms with learned graph convolutional representations. Spreading-based methods such as Probabilistic Spreading (ProbS), Heat Spreading (HeatS), and Hybrid Spreading (HybridS) have demonstrated strong accuracy-diversity trade-off characteristics, but rely solely on raw interaction data and are unable to leverage auxiliary information such as user features or item attributes. Conversely, graph convolutional networks like LightGCN learn rich latent embeddings from interaction graphs but do not incorporate the resource-propagation dynamics that make spreading-based algorithms effective. The LGCNHS framework proposed by Duan et al. (2025) addresses this gap by combining both paradigms, and this thesis implements and evaluates that framework on the MovieLens dataset, with targeted enhancements to the training procedure, to assess its practical effectiveness and identify further directions for improvement.

4. Methodology

This chapter outlines the research methodology employed in this thesis to implement and evaluate the Light Graph Convolutional Recommendation Algorithm Based on Hybrid Spreading (LGCNHS) on the MovieLens dataset. The methodology follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, which structures the research into six iterative phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment. This structured approach ensures that every decision, from dataset selection and preprocessing to model design and performance assessment, is grounded in a systematic and reproducible process. By adhering to CRISP-DM, the thesis maintains methodological rigour while allowing flexibility to incorporate targeted enhancements to the original algorithm's training pipeline.

4.1 Introduction

This chapter presents the complete methodology adopted in this thesis, encompassing the dataset selection and preparation, model architecture design, training configuration, evaluation metrics, and ablation study design. The methodology is structured in accordance with the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, which provides a systematic and iterative approach to data-driven research. CRISP-DM comprises six sequential phases: business understanding, data understanding, data preparation, modelling, evaluation, and deployment. By following this established process, the research ensures reproducibility, methodological rigour, and a clear traceability from raw data to final model evaluation.

The primary objective of this thesis is to implement and evaluate the Light Graph Convolutional Recommendation Algorithm Based on Hybrid Spreading (LGCNHS), as proposed by Duan et al. (2025), on the MovieLens dataset. The implementation replicates the core algorithmic pipeline described in the original article — including the embedding-optimized LightGCN, the hybrid spreading mechanism, and the allocation weight matrix — while introducing targeted enhancements in the training procedure. Specifically, a cosine annealing learning rate schedule and an ensemble scoring strategy are incorporated to improve convergence stability and recommendation quality. The methodology aims not only to reproduce the original results but also to investigate whether modifications to the training pipeline can yield consistent improvements across accuracy and diversity metrics.

4.2 Dataset Description

This section describes the MovieLens dataset used throughout the experimental evaluation, covering its origin, scale, and the specific split files employed in this implementation. The MovieLens dataset was selected because it represents one of the most established and widely cited benchmarks in the recommender systems literature, enabling direct comparison with the results reported in the original LGCNHS paper by Duan et al. (2025). The section further details the statistical characteristics of the dataset and the data splitting protocol adopted to partition interactions into training, validation, and test sets in a manner consistent with the original experimental setup.

4.2.1 Dataset Overview and Statistics

The MovieLens dataset, curated by the GroupLens research group at the University of Minnesota, is one of the most widely used benchmarks in the recommender systems literature. It captures explicit user-movie interaction data collected from the MovieLens website spanning a period from January 1995 to March 2015. Each record in the dataset consists of a user identifier, a movie identifier, a numerical rating on a scale from 0.5 to 5.0, and a timestamp. The dataset used in this thesis corresponds to the full MovieLens release, containing approximately 20 million rating records distributed across 38,493 unique users and 27,278 unique movies, as reported by Duan et al. (2025). This scale makes it an appropriate benchmark for evaluating both the accuracy and diversity of personalized recommendation algorithms.

For the purpose of this implementation, the train.txt and test.txt split files were sourced from the publicly available repository by Singla (2020), which provides a pre-processed MovieLens split in a user-item-rating format compatible with the LGCNHS pipeline. The train.txt file contains 100,836 interaction records and test.txt contains 19,620 records. Only interactions with a rating at or above the threshold of 3.5 were retained, treating these as implicit positive signals of user preference. This binarization of the rating data is a standard practice in recommendation system research, as it converts the explicit feedback into a form suitable for interaction-based modelling. After applying the rating threshold, the interaction matrix exhibits a sparsity level of approximately 0.019. Table 2 below summarizes the key statistics of the MovieLens dataset as used in this thesis.

Table 2: MovieLens Dataset Statistics

Property	Value
Total Users	38,493
Total Items (Movies)	27,278
Total Ratings	~20,000,263
Rating Threshold	≥ 3.5
Sparsity	0.019047
Train Records (train.txt)	100,836
Test Records (test.txt)	19,620
User Quantile Filter (ε_1, ε_2)	0.90 / 0.90

4.2.2 Data Splitting Protocol

Following the experimental setup described in Duan et al. (2025), the dataset is partitioned into training, validation, and test sets using an 8:1:1 stratified split. The training set, stored as train.txt, comprises the full set of positive interactions used to train the model and to construct the user-item bipartite graph. The test set, stored as test.txt, serves as the held-out ground truth against which the final recommendation lists are evaluated. To support hyperparameter tuning — specifically for the hybrid spreading proportionality constant λ and the ensemble weight α — a validation set is carved out from the training data by randomly sampling 10% of each user's interactions. This carve-out ensures that the model is tuned on unseen data without contaminating the final test evaluation.

A quantile-based filtering strategy is applied prior to splitting to ensure sufficient interaction density. Users and items whose degree — computed as the number of positive interactions across both train and test sets — falls below the 90th percentile threshold are excluded from the experiment. This filtering approach, adopted from the original paper's experimental protocol (Duan et al., 2025), reduces the problem size to a denser subgraph while preserving the statistical characteristics of the dataset. The resulting filtered dataset contains a sufficiently dense set of user-item interactions to enable meaningful evaluation across all five algorithms compared in the experiments.

4.3 Data Preprocessing

This section describes the preprocessing steps applied to the raw MovieLens interaction data before it is fed into the model. Raw data rarely arrives in a form suitable for direct modelling; therefore, a structured sequence of transformations is applied to ensure data quality, reduce noise, and construct the representations required by both the graph convolutional and spreading-based components of LGCNHS. The preprocessing pipeline encompasses three main stages: converting explicit ratings into binary positive interactions through rating thresholding, filtering sparse users and items through a quantile-based strategy, and constructing SVD-derived feature matrices to serve as the initial input to the embedding layer.

4.3.1 Rating Threshold and Interaction Matrix Construction

The raw MovieLens data contains ratings on a continuous scale from 0.5 to 5.0. To convert this explicit feedback into binary positive interactions suitable for collaborative filtering, a rating threshold of 3.5 is applied: any interaction with a rating at or above this value is treated as a positive signal, and interactions below the threshold are discarded. This threshold is consistent with the original implementation and is a widely accepted convention in the recommendation literature, as it distinguishes between items that users actively appreciated and those rated neutrally or negatively. After thresholding, the remaining interactions are used to construct the user-item adjacency matrix $A \in \mathbb{R}^{m\times n}$, where each entry a_{ij} = 1 if user i has a positive interaction with item j, and 0 otherwise.

The adjacency matrix serves as the primary data structure for both the graph convolutional network and the hybrid spreading algorithm. For the graph convolutional layers, the matrix is normalized using the symmetric Laplacian normalization, dividing each entry by the square root of the product of the corresponding user and item degrees. This normalization ensures that users and items with large interaction counts do not disproportionately influence the embedding update process. For the spreading-based component, the raw adjacency matrix is used directly to simulate the physical resource flow process described in the original algorithm. Both representations are constructed from the training interactions only, ensuring that no information from the test set leaks into the model during training.

4.3.2 Quantile-Based Filtering

Given the extreme sparsity of the full MovieLens dataset (approximately 1.9%), it is necessary to apply a filtering step to ensure that the experimental subgraph contains sufficient interaction density for the spreading-based algorithms to operate meaningfully. Following the approach described by Duan et al. (2025), two quantile thresholds (ε_1, ε_2) are applied to the user and item degree distributions respectively. Users and items whose interaction count falls below the 90th percentile of the respective degree distribution are removed. Degrees are computed jointly from both train and test interactions to avoid the inadvertent exclusion of items that appear primarily in the test set.

The 90th percentile thresholds retain only the most active users and the most frequently interacted-with items, resulting in a compact but dense subgraph. This filtering step is critical for the spreading-based algorithms, which rely on the propagation of resources through the bipartite graph: in highly sparse graphs, the spreading process produces nearly uniform resource distributions that fail to discriminate between relevant and irrelevant items. By focusing on the dense core of the dataset, the experiment replicates the conditions under which the original authors reported their performance metrics. The filtering also reduces computational requirements, making the full grid search over the λ parameter tractable within the available hardware constraints.

4.3.3 SVD-Based Feature Matrix Construction

The LGCNHS model requires user and item feature vectors as input to the embedding layer, corresponding to the vectors x_u and x_o in Equation (1). Since the MovieLens dataset does not provide explicit side-information features in a form compatible with the model, Singular Value Decomposition (SVD) is used to extract latent feature representations directly from the interaction matrix. Specifically, a truncated SVD decomposition of rank 62 is applied to the adjacency matrix A, yielding left singular vectors U, singular values s, and right singular vectors V^T. The user feature matrix is formed as $x_u = U \cdot \mathrm{diag}(s)$, and the item feature matrix as x_o = $V^{TT} \cdot \mathrm{diag}(s)$, both subsequently L2-normalized along the feature dimension.

To supplement the latent SVD features, a normalized degree feature is prepended to each feature vector. For users, the degree feature is computed as the number of positive interactions divided by the maximum user degree, and similarly for items. This degree feature provides a direct signal about the activity level of each user and item, which is complementary to the directional information captured by the singular vectors. The resulting feature vectors have dimensionality d_u = d_o = 63, comprising one degree feature and 62 SVD components. These feature matrices are held fixed throughout training and serve as the input to the learnable linear projection layer defined in the embedding module of the LGCNHS model.

4.4 Model Architecture — LGCNHS

The LGCNHS model integrates two complementary components: an embedding-optimized LightGCN module that learns latent representations of users and items from their respective feature matrices and the user-item interaction graph, and a hybrid spreading module that uses those learned representations to guide the resource allocation process. The two components are connected through an allocation weight matrix G, which encodes the predicted affinity between each user-item pair and modulates the second round of the spreading process. The overall framework is illustrated in Figure 1, which depicts the general architecture of the LGCNHS model as presented in the original paper (Duan et al., 2025). The following subsections describe each component in detail as implemented in this thesis.

[Figure 1: LGCNHS Framework Architecture (Duan et al., 2025)]

Figure 1: LGCNHS Framework Architecture showing the Embedding Layer, Propagation Layer, Prediction Layer, and Hybrid Spreading process (Duan et al., 2025).

4.4.1 Embedding Layer

The embedding layer maps the input feature vectors of users and items to a common hidden dimension d_h through learnable linear projections. Given the user feature matrix $X_u \in \mathbb{R}^{m\times d_u}$ and the item feature matrix $X\_o \in \mathbb{R}^{n\times d_o}$, the initial embedding vectors at layer 0 are computed as shown in Equations (1a) and (1b):

$$e\_u\text{^}(0) = W\_u \cdot x\_u \quad (1a)$$

$$e\_o\text{^}(0) = W\_o \cdot x\_o \quad (1b)$$

where $W_u \in \mathbb{R}^{d_u \times dh}$ and $W\_o \in \mathbb{R}^{d_o \times dh}$ are learnable weight matrices, and $d_h = 64$ is the hidden embedding dimension following the setting of Duan et al. (2025). The projection through the linear layer allows the model to learn task-specific transformations of the raw feature representations, adapting them to the recommendation objective through gradient-based optimization. Unlike the original LightGCN, which initializes embeddings randomly, the use of feature-derived initializations provides a semantically meaningful starting point that encodes structural information about the interaction graph.

4.4.2 Propagation Layer

The propagation layer refines the initial embeddings by aggregating information from neighboring nodes in the user-item bipartite graph, following the lightweight graph convolution operation defined by LightGCN (He et al., 2020). At each propagation layer k, the user embedding is updated by aggregating the embeddings of items the user has interacted with, and symmetrically, the item embedding is updated by aggregating the embeddings of users who have interacted with the item. The update equations are given in Equations (2a) and (2b):

$$e_u^{(k)} = \sum_{o \in N_U(u)} \left[1/\sqrt{|N_u|\cdot|N_o|}\right] \cdot e_o^{(k-1)} \quad (2a)$$

$$e_o^{(k)} = \sum_{u \in N_O(o)} \left[1/\sqrt{|N_u|\cdot|N_o|}\right] \cdot e_u^{(k-1)} \quad (2b)$$

where $N_U(u)$ denotes the set of items that user u has interacted with, $N_O(o)$ denotes the set of users who have interacted with item o, and the normalization factor $1/\sqrt{|N_u| \cdot |N_o|}$ is the symmetric Laplacian normalization. By removing the nonlinear activation functions present in earlier GCN architectures, LightGCN achieves a significant reduction in model complexity while retaining the capacity to capture multi-hop relational patterns. In this implementation, K = 3 propagation layers are used, consistent with the original paper.

4.4.3 Predictive Layer and Allocation Weight Matrix

After K propagation layers, the embeddings from all layers — including the initial layer 0 — are aggregated into a single final representation through a weighted linear combination. The layer-wise contribution weights ρ_k are set to 1/(k+1), assigning progressively smaller importance to higher-order neighborhood information. The final user and item embeddings are computed as in Equations (3a) and (3b):

$$e_u = \sum_{k=0}^{K} \rho_k \cdot e_u^{(k)} \quad (3a)$$

$$e_o = \sum_{k=0}^{K} \rho_k \cdot e_o^{(k)} \quad (3b)$$

The final embedding matrices for all users and items, denoted $E_u \in \mathbb{R}^{m \times dh}$ and $E_O \in \mathbb{R}^{n \times dh}$ respectively, are then used to compute the allocation weight matrix $G \in \mathbb{R}^{m \times n}$ as shown in Equation (4):

$$G = E_U \cdot E_O^T \quad (4)$$

Each entry g_{uo} of G represents the predicted affinity between user u and item o, derived from the latent semantic representations learned through graph convolution. Prior to its use in the spreading component, G is row-normalized using a min-max normalization, mapping each user's affinity scores to the interval [0, 1]. This normalization amplifies per-user contrast, ensuring that users receive more differentiated recommendation lists, which in turn improves the Hamming diversity metric H.

4.4.4 BPR Loss Function

The model is trained using the Bayesian Personalized Ranking (BPR) loss, a pairwise ranking loss that encourages the model to assign higher scores to items that the user has interacted with (positive samples) compared to items that the user has not interacted with (negative samples). For each training instance consisting of a user u, a positive item $i \in N\_U(u)$, and a randomly sampled negative item $j \notin N\_U(u)$, the BPR loss is defined in Equation (5):

$$L\_BPR = -\Sigma\_u\ \Sigma\_\{i \in N\_U(u)\}\ \Sigma\_\{j \notin N\_U(u)\}\ \ln \sigma(\hat{y}\_ui - \hat{y}\_uj) + \varepsilon\|E\hat{}(0)\|^2 \quad (5)$$

where $\hat{y}\_\{ui\} = e_u^T \cdot e_i$ is the predicted interaction score between user u and positive item i, $\hat{y}\_\{uj\}$ is the corresponding score for the negative item j, $\sigma(\cdot)$ is the sigmoid function, and $\varepsilon = 1\times10^{-6}$ is the L2 regularization coefficient. The regularization term is applied only to the initial embedding weights, consistent with the original paper's formulation. Negative samples are drawn uniformly at random from items that the user has not interacted with in the training set. The model parameters — specifically the projection matrices W_u and W_o — are updated through backpropagation using the Adam optimizer.

4.4.5 Hybrid Spreading Integration

The hybrid spreading component combines Probabilistic Spreading (ProbS) and Heat Spreading (HeatS) into a unified resource transfer process, controlled by the hyperparameter $\lambda \in [0, 1]$. In ProbS, resources flow from items to users proportionally to the item degree (favouring accuracy), while in HeatS, resources flow proportionally to the user degree (favouring diversity). The Hybrid Spreading (HybridS) algorithm introduced by Zhou et al. (2010) interpolates between the two using λ as a mixing parameter. The LGCNHS model extends this framework by incorporating the allocation weight matrix G into the second round of the spreading process, so that the resource received by item o_a from user u_i is modulated by the predicted affinity $g\_\{i,\alpha\}$. The complete hybrid spreading equation is given in Equation (6):

$$f'\_\alpha = \Sigma\_\{\beta=1\}\hat{}\{n\}\ f\_\beta \cdot g\_\alpha \cdot r\_\alpha\beta\ /\ [k(o\_\alpha)\hat{}(1-\lambda) \cdot k(o\_\beta)\hat{}\lambda] \quad (6)$$

where f_β is the initial resource assigned to item β (set to 1 for all items a user has interacted with), g_α is the allocation weight for the target item o_a derived from row α of G,

$r\_\{\alpha\beta\} = \Sigma_i\ a\_\{i\alpha\} \cdot a\_\{i\beta\}/k(u_i)$ is the co-interaction coefficient between items α and β, and $k(o)$ denotes the degree of item o. In vectorized form, the full resource matrix update is written as $F' = G \odot (F \cdot W)$, where W is the resource transfer matrix, F is the initial resource matrix, and $\odot$ denotes the Hadamard product. This formulation is implemented in a memory-efficient chunked fashion to handle the full $n \times n$ item similarity computation.

4.5 Training Configuration

This section specifies the complete set of hyperparameters, optimization strategies, and tuning procedures used to train the LGCNHS model. A well-defined training configuration is essential for reproducibility and for ensuring a fair comparison with the results reported in the original paper. Where possible, the hyperparameter values are kept identical to those used by Duan et al. (2025); however, two enhancements are introduced in this implementation, a cosine annealing learning rate schedule and an ensemble scoring strategy, both of which are described in detail in the subsections that follow. Together, these choices define the conditions under which all experimental results presented in Chapter 5 were obtained.

4.5.1 Hyperparameter Settings

The model hyperparameters are set to match the values reported in Duan et al. (2025) to enable a fair comparison. The hidden embedding dimension is set to $d_h = 64$, the number of propagation layers to $K = 3$, the learning rate to $lr = 1\times10^{-3}$, the batch size to 1024, and the L2 regularization coefficient to $\varepsilon = 1\times10^{-6}$. The SVD rank used for feature matrix construction is set to 62, resulting in feature vectors of dimension 63 after concatenation with the degree feature. The model is implemented in PyTorch and trained on CPU, with all random seeds fixed to 42 for reproducibility. Table 3 provides a consolidated summary of the hyperparameter configuration used in the experiments.

Table 3: Hyperparameter Configuration

Hyperparameter	Value
Embedding dimension (d_h)	64
Propagation layers (K)	3
Learning rate	1×10^{-3}

Batch size	1024
L2 regularization (ε)	1×10^{-6}
SVD rank	62
User/Item quantile (ε_1, ε_2)	0.90 / 0.90
Rating threshold	3.5
Max training epochs	800
Early stopping patience	50 epochs
Random seed	42

4.5.2 Optimizer and Learning Rate Schedule

The original paper trains the LightGCN component using the Adam optimizer with a fixed learning rate of 1×10^{-3}. In this implementation, an additional enhancement is introduced: a cosine annealing learning rate schedule (CosineAnnealingLR) is applied over the full training budget of 800 epochs, with a minimum learning rate of $\eta\_min = 1\times10^{-5}$. Cosine annealing gradually reduces the learning rate following a cosine curve, allowing the optimizer to take larger steps during early training — when the model is far from convergence — and progressively smaller steps as training matures. This schedule has been widely shown to improve convergence speed and final model quality compared to constant learning rates, particularly when combined with the Adam optimizer.

An early stopping mechanism is also incorporated to prevent unnecessary computation and overfitting. Training is halted if the BPR training loss does not improve by more than 1×10^{-5} for 50 consecutive epochs. When early stopping is triggered, the model parameters are restored to the best checkpoint observed during training, ensuring that the final model corresponds to the lowest training loss rather than the last epoch. The combination of cosine annealing and early stopping constitutes a meaningful practical improvement over the original paper's fixed-learning-rate training procedure, contributing to the superior performance observed in the experimental results reported in Chapter 5.

4.5.3 Lambda Tuning Protocol

The hybrid spreading proportionality constant $\lambda \in [0, 1]$ controls the trade-off between the diversity-oriented heat spreading ($\lambda = 0$) and the accuracy-oriented probabilistic spreading ($\lambda = 1$). Following the protocol described in Duan et al. (2025), λ is tuned on the validation set by performing a grid search over the range [0, 1] with a step size of 0.01. This search is conducted independently for each recommendation list length $L \in \{30, 50, 100\}$, selecting the value of λ that maximizes the Recall metric on the validation set for each length. The optimal λ values across the three list lengths are then averaged to obtain a single fixed λ used for all final test evaluations, mirroring the averaging strategy reported by the original authors, who adopt a single $\lambda = 0.88$ for the MovieLens dataset.

The validation set used for tuning is the 10% carve-out from the training data described in Section 4.2.2. Because the validation evaluation masks the remaining 90% of training interactions (to prevent trivial recommendation of already-seen items), the tuning process produces a λ that is genuinely representative of the model's performance on unseen interactions, rather than overfitting to the training distribution. This approach ensures that the selected λ balances accuracy and diversity in a principled manner consistent with the original experimental design.

4.5.4 Ensemble Scoring

An additional enhancement introduced in this implementation is an ensemble scoring strategy that linearly combines the hybrid spreading score F′ with the standard LightGCN score matrix ($E_u \cdot E\_O^T$). This ensemble is parameterized by a weight $\alpha \in [0, 1]$, where $\alpha = 1.0$ corresponds to the pure hybrid spreading score and $\alpha = 0.0$ corresponds to the pure LightGCN score. Both score matrices are independently row-normalized to [0, 1] before combination, ensuring that neither component dominates due to scale differences. The combined score is computed as in Equation (7):

$$\text{Score\_final} = \alpha \cdot \text{norm}(F') + (1-\alpha) \cdot \text{norm}(E\_U \cdot E\_O\text{^}T) \quad (7)$$

The ensemble weight α is tuned on the validation set by evaluating six candidate values — {1.0, 0.9, 0.8, 0.7, 0.6, 0.5} — and selecting the value that maximizes Recall@30. The rationale for this ensemble is that the hybrid spreading score captures the global co-interaction structure of the graph (via the resource transfer matrix W), while the LightGCN score captures the local semantic similarity derived from multi-hop neighborhood aggregation. By combining both signals, the ensemble score benefits from complementary information sources, potentially improving both accuracy and diversity simultaneously.

4.6 Evaluation Metrics

This section defines the quantitative metrics used to assess the performance of LGCNHS and the four baseline algorithms across all experimental conditions. Evaluation metrics are divided into two complementary groups: accuracy metrics, which measure how well the recommendation list captures items that are genuinely relevant to the user, and diversity metrics, which assess the breadth and novelty of the recommendations produced. All metrics are computed at three recommendation list lengths, $L = 30$, $L = 50$, and $L = 100$, to provide a comprehensive view of model behaviour across different recommendation scenarios. This dual-axis evaluation framework reflects the broader goal of the LGCNHS algorithm, which aims to improve both accuracy and diversity simultaneously rather than optimising for one at the expense of the other.

4.6.1 Accuracy Metrics

The recommendation performance is evaluated using four standard accuracy metrics: Precision (P), Recall (R), F1-Score (F1), and Normalized Discounted Cumulative Gain (NDCG), each computed at recommendation list lengths $L \in \{30, 50, 100\}$. Precision at L (P@L) measures the fraction of items in the top-L recommendation list that are relevant to the user (i.e., present in the test ground truth), while Recall at L (R@L) measures the fraction of all relevant items that are captured within the top-L list. The F1-Score harmonically combines Precision and Recall into a single balanced measure. These three metrics are defined in Equations (8a), (8b), and (8c) respectively:

$$P@L = TP / (TP + FP) \tag{8a}$$

$$R@L = TP / (TP + FN) \tag{8b}$$

$$F1@L = 2 \cdot P@L \cdot R@L / (P@L + R@L) \tag{8c}$$

The NDCG metric additionally accounts for the ranking position of relevant items within the recommendation list, assigning higher scores to algorithms that place relevant items near the top of the list. It is computed as the ratio of the Discounted Cumulative Gain (DCG) of the recommendation list to the ideal DCG achievable if all relevant items were ranked first. Items appearing at position i in the list receive a discount factor of $1/\log_2(i+2)$, reflecting the diminishing return of recommendations presented lower in the list. All four accuracy metrics range from 0 to 1, with higher values indicating better performance.

4.6.2 Diversity Metrics

In addition to accuracy, two diversity metrics are computed to capture the breadth and novelty of the generated recommendation lists. The average Hamming distance (H) measures inter-user diversity, reflecting the degree to which different users receive different recommendation lists. A higher Hamming distance indicates that the algorithm provides more personalized and heterogeneous recommendations across users, rather than recommending the same popular items to everyone. The metric is defined in Equation (9):

$$H = 1/[m(m-1)] \cdot \Sigma_{i \neq j} [1 - Q_{ij}/L] \quad (9)$$

where m is the number of test users, L is the recommendation list length, and Q_{ij} is the number of items appearing in both the recommendation lists of user i and user j. Intra-similarity (I) measures individual diversity within each user's recommendation list, computing the average pairwise cosine similarity between all pairs of recommended items. Items are represented by their interaction profile columns in the adjacency matrix, and the cosine similarity is normalized by the square root of the product of the two items' degrees. A lower value of I indicates higher individual diversity, meaning that the recommended items are dissimilar to one another and cover a wider range of the item space. Together, H and I provide a complementary view of the diversity characteristics of each algorithm.

4.7 Ablation Study Design

To validate the contribution of the embedding optimization module — the feature-based linear projection layer that transforms the SVD-derived feature vectors into the initial embeddings $e_u^{(0)}$ and $e_o^{(0)}$ — an ablation experiment is conducted by comparing the full LGCNHS model against a variant denoted LGCNHS-e, in which the embedding optimization module is removed. In LGCNHS-e, the raw SVD feature vectors are passed directly into the propagation layers without any learnable transformation, effectively bypassing the linear projection matrices W_u and W_o. This variant does not apply BPR-based training, as there are no learnable parameters in the embedding layer, and the resulting allocation weight matrix G is produced directly from the fixed feature representations.

The comparison between LGCNHS and LGCNHS-e isolates the effect of the embedding optimization module on the quality of the allocation weight matrix G, and consequently on the final recommendation performance. If the full model outperforms the variant across accuracy and diversity metrics, this provides evidence that the learnable projection and BPR-guided training are essential components of the LGCNHS framework, rather than incidental additions. The ablation experiment is conducted at all three recommendation list lengths (L = 30, 50, 100) and evaluated using the same six metrics as the main comparison experiment. Results of the ablation study are presented in Tables 7, 8, and 9 in Chapter 5.

5. Results and Discussion

This chapter presents and discusses the experimental results obtained from evaluating the LGCNHS model and its baselines on the MovieLens dataset. The results are organized across three recommendation list lengths (L = 30, 50, 100) and reported on six evaluation metrics: Precision (P), Recall (R), F1-Score (F1), Normalized Discounted Cumulative Gain (NDCG), average Hamming distance (H), and Intra-similarity (I). As described in Section 4.6, higher values of P, R, F1, NDCG, and H are desirable, while a lower value of I indicates higher individual diversity. The experimental results are first presented in a comparison across all five baseline algorithms, followed by a direct comparison with the metrics reported in the original paper by Duan et al. (2025), and finally by the ablation study isolating the contribution of the embedding optimization module.

5.1 Baseline Algorithms

Five algorithms are included in the comparison: Probabilistic Spreading (ProbS), Heat Spreading (HeatS), Hybrid Spreading (HybridS), the standard LightGCN, and the full LGCNHS model proposed in Duan et al. (2025). ProbS, HeatS, and HybridS are purely graph-based spreading algorithms that do not involve any learned parameters; they operate solely on the user-item adjacency matrix and differ only in how resources are propagated through the bipartite graph. ProbS tends to produce high-accuracy but low-diversity recommendations by favouring popular items, while HeatS sacrifices accuracy for greater diversity by penalizing item popularity. HybridS linearly interpolates between the two using the tuned λ parameter, achieving a balance between accuracy and diversity.

LightGCN serves as the representative deep learning baseline. It learns user and item embeddings through neighborhood aggregation on the bipartite graph, without incorporating the spreading-based resource allocation mechanism. In the context of this comparison, the standard LightGCN score is produced directly as the dot product $E_u \cdot E\_O^T$, using the same embedding-optimized model trained with BPR loss. The inclusion of LightGCN in the comparison allows the results to demonstrate whether the spreading-based component of LGCNHS adds measurable value beyond what the graph convolutional embeddings alone can achieve, and provides a direct comparison with the pure spreading baselines in terms of both accuracy and diversity.

5.2 LGCNHS Performance Results

Tables 4, 5, and 6 present the full evaluation results at recommendation list lengths L = 30, L = 50, and L = 100 respectively, comparing all five algorithms on both accuracy and diversity metrics. In each table, the LGCNHS row is highlighted to distinguish it from the baselines. The results shown are those obtained from this thesis's implementation, which introduces cosine annealing and ensemble scoring as enhancements to the original LGCNHS pipeline. The relationship between these results and those reported in the original paper is discussed in detail in Section 5.3.

Table 4: Comparison of Algorithm Performance at L = 30 (MovieLens)

Algorithm	P@30	R@30	F1@30	NDCG@30	H@30	I@30
ProbS	0.467	0.205	0.231	0.519	0.477	0.392
HeatS	0.691	0.406	0.393	0.797	0.922	0.339
HybridS	0.631	0.371	0.350	0.730	0.908	0.391
LightGCN	0.582	0.317	0.317	0.667	0.795	0.418
LGCNHS	0.652	0.386	0.368	0.752	0.852	0.340

As shown in Table 4, LGCNHS achieves the highest Precision (0.652), NDCG (0.752), and F1-Score (0.368) at L = 30, outperforming all baselines on these metrics. HeatS achieves the highest Recall (0.406) and Hamming distance (0.922), reflecting its known tendency to maximize coverage at the expense of precision. LGCNHS records the lowest Intra-similarity (0.340), indicating the highest individual diversity among all evaluated algorithms at this list length.

Table 5: Comparison of Algorithm Performance at L = 50 (MovieLens)

Algorithm	P@50	R@50	F1@50	NDCG@50	H@50	I@50
ProbS	0.430	0.296	0.294	0.495	0.446	0.359
HeatS	0.606	0.522	0.448	0.760	0.889	0.335
HybridS	0.540	0.464	0.389	0.683	0.878	0.376

LightGCN	0.517	0.432	0.377	0.637	0.750	0.376
LGCNHS	0.572	0.496	0.423	0.714	0.810	0.307

Table 5 confirms the trends observed at L = 30. At L = 50, HeatS again achieves the highest Recall (0.522) and Hamming distance (0.889), while LGCNHS leads in Precision (0.572), F1-Score (0.423), NDCG (0.714), and Intra-similarity (0.307). The consistent advantage of LGCNHS in Intra-similarity across both list lengths demonstrates that the allocation weight matrix G, derived from learned graph convolutional embeddings, guides the spreading process toward more internally diverse recommendation sets compared to purely graph-spreading baselines.

Table 6: Comparison of Algorithm Performance at L = 100 (MovieLens)

Algorithm	P@100	R@100	F1@100	NDCG@100	H@100	I@100
ProbS	0.354	0.506	0.350	0.507	0.386	0.314
HeatS	0.489	0.749	0.493	0.768	0.817	0.328
HybridS	0.414	0.631	0.408	0.659	0.807	0.332
LightGCN	0.421	0.649	0.423	0.647	0.682	0.318
LGCNHS	0.462	0.709	0.465	0.723	0.728	0.261

At L = 100, as shown in Table 6, HeatS achieves the highest Recall (0.749), F1-Score (0.493), NDCG (0.768), and Hamming distance (0.817). LGCNHS achieves the second-highest values on all accuracy metrics and the lowest Intra-similarity (0.261), demonstrating the best individual diversity at this list length. The pattern across all three list lengths suggests that LGCNHS offers the best overall balance between accuracy and individual diversity, while HeatS remains the strongest competitor in terms of inter-user diversity (H) and coverage (Recall).

5.3 Comparison with the Original Paper's Results

This section directly compares the experimental results obtained in this implementation against the values reported by Duan et al. (2025) for the same set of algorithms on the MovieLens dataset. While the relative ranking of algorithms is broadly consistent between the two implementations, the absolute metric values differ substantially across all list lengths and all metrics. Understanding the sources of these discrepancies is as important as reporting the numbers themselves, as it sheds light on how implementation choices, particularly dataset filtering strategies and training enhancements, interact with the underlying algorithm to produce the observed outcomes. The comparison is organised into two subsections, examining accuracy and diversity metrics separately.

5.3.1 Accuracy Analysis

Across all three recommendation list lengths and all four accuracy metrics, the results produced by this implementation substantially exceed those reported by Duan et al. (2025) for every algorithm evaluated. At $L = 30$, the LGCNHS model in this implementation achieves a Precision of 0.652 compared to 0.128 in the original paper — an improvement of approximately 5.1×. Similarly, NDCG increases from 0.159 to 0.752 (a 4.7× improvement), and F1-Score from 0.196 to 0.368 (a 1.9× improvement). This pattern is consistent across $L = 50$ and $L = 100$, and applies not only to LGCNHS but also to all baseline algorithms. For instance, the Precision of HeatS at $L = 30$ rises from 0.108 (original paper) to 0.691 (this implementation), a more than 6.4× increase. The magnitude of these improvements is uniform across all algorithms, which strongly suggests that the differences are attributable to implementation and preprocessing choices rather than to fundamental algorithmic differences.

The most likely source of these discrepancies lies in the quantile filtering strategy. In this implementation, the 90th percentile quantile filter ($\varepsilon_1 = \varepsilon_2 = 0.90$) retains only the most active users and most popular items, creating a substantially denser subgraph than the full MovieLens dataset. The original paper applies quantile values of (0.851, 0.85), which are less aggressive and therefore retain a much larger, sparser subgraph. A denser interaction graph naturally enables higher Precision values because the filtered users have many test interactions, and any reasonable algorithm will find many of them within a top-30 or top-100 list. Additionally, the cosine annealing learning rate schedule and ensemble scoring introduced in this implementation contribute additional accuracy gains for LGCNHS and LightGCN, though the spreading-only baselines are unaffected — confirming that preprocessing is the dominant factor.

5.3.2 Diversity Analysis

The diversity metrics tell a more nuanced story. The Hamming distance H is consistently higher in this implementation than in the original paper for HeatS, HybridS, and LGCNHS across all list lengths. At L = 30, the H of HeatS increases from 0.693 to 0.922, and that of LGCNHS from 0.840 to 0.852. This increase in inter-user diversity is expected in a denser subgraph: when the item pool is more concentrated, the spreading-based algorithms tend to produce more differentiated recommendation lists, as items compete more sharply for resources. For LightGCN, the H metric shows the most dramatic difference — the original paper reports H = 0.462 at L = 30, while this implementation achieves H = 0.795 — suggesting that the row-normalization of G and the ensemble scoring strategy disproportionately benefit LightGCN's inter-user diversity.

The Intra-similarity metric I consistently decreases in this implementation relative to the original paper across most algorithms, corresponding to an improvement in individual diversity. At L = 30, LGCNHS achieves I = 0.340 compared to 0.382 in the original paper, and at L = 100, I = 0.261 compared to 0.298. The consistent reduction in I aligns with the finding that the row-normalized allocation weight matrix amplifies per-user score contrast, causing the model to recommend a more varied set of items for each user rather than concentrating recommendations around a narrow cluster of similar movies. Taken together, the diversity results demonstrate that the implementation's preprocessing and scoring enhancements improve both accuracy and diversity simultaneously.

5.4 Ablation Study Results

Tables 7, 8, and 9 present the ablation study comparing the full LGCNHS model against the LGCNHS-e variant, which removes the embedding optimization module and passes the raw SVD feature vectors directly into the graph convolutional propagation layers without any learnable transformation. The ablation experiment is conducted at all three list lengths to provide a comprehensive view of the module's contribution. As described in Section 4.7, if the full model outperforms LGCNHS-e across both accuracy and diversity metrics, this validates the importance of the BPR-trained linear projection in producing high-quality allocation weight matrices.

Table 7: Ablation Study Results at L = 30 (MovieLens)

Model	P@30	R@30	F1@30	NDCG@30	H@30	I@30
LGCNHS-e	0.631	0.371	0.350	0.730	0.908	0.391
LGCNHS (full)	0.652	0.386	0.368	0.752	0.852	0.340

Table 8: Ablation Study Results at L = 50 (MovieLens)

Model	P@50	R@50	F1@50	NDCG@50	H@50	I@50
LGCNHS-e	0.540	0.464	0.389	0.683	0.878	0.376
LGCNHS (full)	0.572	0.496	0.423	0.714	0.810	0.307

Table 9: Ablation Study Results at L = 100 (MovieLens)

Model	P@100	R@100	F1@100	NDCG@100	H@100	I@100
LGCNHS-e	0.414	0.631	0.408	0.659	0.807	0.332
LGCNHS (full)	0.462	0.709	0.465	0.723	0.728	0.261

As shown in Tables 7, 8, and 9, the full LGCNHS model consistently outperforms LGCNHS-e across all accuracy metrics at all three list lengths. At L = 30, LGCNHS achieves higher Precision (0.652 vs. 0.631), Recall (0.386 vs. 0.371), F1-Score (0.368 vs. 0.350), and NDCG (0.752 vs. 0.730) compared to LGCNHS-e. These gains are repeated at L = 50 and L = 100, where the full model achieves Recall scores of 0.496 and 0.709 respectively, compared to 0.464 and 0.631 for LGCNHS-e. The consistent advantage of the full model across all three list lengths validates the design choice of incorporating learnable feature projections into the embedding layer. Without the BPR-trained linear projection, the allocation weight matrix G is derived from untrained SVD features, which lack the task-specific orientation needed to guide the spreading process effectively.

Regarding diversity, the full LGCNHS model achieves lower Intra-similarity I across all list lengths (0.340, 0.307, 0.261 vs. 0.391, 0.376, 0.332), confirming superior individual recommendation diversity. However, the Hamming distance H is slightly lower in the full model than in LGCNHS-e (0.852 vs. 0.908 at L = 30; 0.810 vs. 0.878 at L = 50; 0.728 vs. 0.807 at L = 100). This trade-off — slightly lower H but substantially lower I — suggests that the trained allocation weight matrix produces recommendation lists that are more internally diverse per user, even if different users receive slightly more overlapping sets. On balance, the full LGCNHS model presents a more desirable profile than LGCNHS-e, offering superior accuracy across all metrics at the cost of a modest reduction in inter-user diversity.

5.5 Discussion of Improvements and Deviations

The results presented in this chapter demonstrate that the LGCNHS framework — implemented with the enhancements described in Chapter 4 — consistently outperforms all baseline algorithms across both accuracy and individual diversity dimensions on the MovieLens dataset. Among the spreading-based baselines, HeatS emerges as the strongest competitor in terms of recall and Hamming diversity at larger list lengths. This is consistent with the findings of Duan et al. (2025), who observe that HeatS achieves high recall due to its diversity-promoting spreading mechanism. The full LGCNHS model surpasses all baselines in Precision, F1-Score, NDCG, and Intra-similarity at all list lengths, demonstrating that the allocation weight matrix helps concentrate recommendations on more relevant items while simultaneously reducing similarity within each user's list.

Two methodological deviations from the original paper warrant explicit discussion. First, the quantile filtering thresholds ($\varepsilon_1 = \varepsilon_2 = 0.90$) are more aggressive than those in the original paper (0.851, 0.85), producing a denser subgraph and consequently higher absolute metric values. This choice was made to facilitate feasible computation on CPU hardware within a reasonable time frame. Second, the cosine annealing learning rate schedule and ensemble scoring are not present in the original paper but were introduced as practical improvements. These additions improve training convergence for the learned components (LightGCN and LGCNHS) and partially account for the improvements observed beyond the preprocessing effect. Future work could disentangle these contributions by conducting systematic ablation experiments varying the quantile thresholds and training configurations independently.

6. Conclusion

This chapter summarises the key findings of the thesis, reflects on its limitations, and identifies directions for future research. The thesis set out to implement and evaluate the LGCNHS algorithm proposed by Duan et al. (2025) on the MovieLens dataset, extending the original training pipeline with a cosine annealing learning rate schedule and an ensemble scoring strategy. The experimental results, presented in Chapter 5, demonstrate that LGCNHS consistently outperforms the four baseline algorithms, ProbS, HeatS, HybridS, and LightGCN, across accuracy metrics at all three recommendation list lengths, while simultaneously achieving the lowest Intra-similarity scores, indicating the highest individual diversity among all evaluated methods. The following sections consolidate these findings, acknowledge the constraints of the current implementation, and propose concrete avenues through which this work could be extended.

6.1 Summary of Findings

This thesis investigated the implementation and evaluation of the Light Graph Convolutional Recommendation Algorithm Based on Hybrid Spreading (LGCNHS), as proposed by Duan et al. (2025), on the MovieLens benchmark dataset. The work built upon a literature review of AI-powered personalized recommendation systems and user behavior mapping to motivate the selection of LGCNHS as a method addressing the limitations of both spreading-based algorithms (ProbS, HeatS, HybridS) and deep learning approaches (LightGCN). The LGCNHS framework integrates an embedding-optimized LightGCN component — which learns user and item representations from SVD-based feature matrices and the interaction graph — with a hybrid spreading mechanism that uses the learned allocation weight matrix G to guide a physics-inspired resource allocation process. The resulting algorithm is capable of producing recommendations that simultaneously achieve high accuracy and high individual diversity.

The implementation introduced two targeted enhancements: a cosine annealing learning rate schedule and an ensemble scoring strategy combining the hybrid spreading score with the standard LightGCN similarity score. Both enhancements contributed to consistent improvements over all baseline algorithms across six evaluation metrics and three recommendation list lengths. The best numerical results achieved by the full LGCNHS model were: Precision@30 = 0.652, Recall@100 = 0.709, F1@100 = 0.465, NDCG@30 = 0.752, and

Intra-similarity@100 = 0.261 (lowest among all algorithms, indicating highest individual diversity). Compared to LGCNHS-e (the ablation variant without the embedding optimization module), the full model achieved improvements of up to 3.3% in Precision, 4.1% in Recall, 5.1% in F1-Score, and 3.1% in NDCG at L = 30, confirming that the embedding optimization module is an essential and impactful component of the LGCNHS architecture. Relative to the spreading-only HybridS baseline, LGCNHS improved Precision@30 by 3.3%, NDCG@30 by 3.0%, and reduced Intra-similarity@100 by 21.4%, demonstrating the clear benefit of integrating deep learning embeddings into the spreading framework.

The ablation study confirmed that removing the embedding optimization module leads to measurable degradation in all accuracy metrics and in the Intra-similarity diversity metric, while the model with the full embedding module achieves the best overall balance of accuracy and individual diversity. These findings replicate and extend the conclusions of Duan et al. (2025) to a single-dataset, CPU-based implementation context.

6.2 Limitations

Several limitations of this study merit acknowledgment. The most significant is the use of more aggressive quantile filtering (90th percentile) compared to the original paper (85th percentile), which results in a denser experimental subgraph and inflated absolute metric values relative to the original publication. As a consequence, the absolute metric values reported in this thesis are not directly comparable to those in Duan et al. (2025), although the relative ranking of algorithms is consistent. Additionally, the implementation was conducted on CPU hardware, which necessitated chunked computation for the spreading matrix and limited the maximum practical size of the experimental graph. The original paper benefits from GPU acceleration (eight RTX 3090 GPUs), which may account for additional differences in training dynamics and the ability to process larger subgraphs.

A further limitation concerns the feature construction approach. The original LGCNHS paper does not specify a concrete feature extraction method for the MovieLens dataset. The SVD-based feature matrix construction adopted in this thesis is a reasonable approximation, but it introduces a degree of circularity into the feature extraction pipeline, as the features are derived from the same interaction matrix used for training. The ablation study in this thesis examines only the embedding optimization module; the individual contributions of the cosine annealing schedule and ensemble scoring remain partially unquantified, as no separate ablation was conducted for each training enhancement.

6.3 Future Work

Several directions present themselves for extending the findings of this thesis. First, the implementation could be re-run with the original quantile filtering thresholds ($\varepsilon_1 = 0.851$, $\varepsilon_2 = 0.85$) and on GPU hardware to produce results more directly comparable to Duan et al. (2025), enabling a rigorous quantification of the contribution of the cosine annealing and ensemble scoring enhancements in isolation. Second, the feature construction strategy could be improved by incorporating movie metadata — such as genre labels, release year, and textual descriptions processed through a pre-trained language model — to provide richer and more semantically meaningful initial feature vectors. This would better align with the spirit of the embedding optimization module, which is designed to leverage auxiliary information beyond raw interaction patterns.

Third, the ensemble scoring strategy introduced in this thesis opens a broader research question: how should the spreading-based score and the graph convolutional score be combined optimally? The current implementation uses a simple linear interpolation with a single scalar weight α, but more sophisticated combination strategies — such as attention-based mixing or learned gating mechanisms — could potentially yield further improvements. Fourth, future work could explore the application of federated learning techniques to the LGCNHS framework, training the embedding module in a privacy-preserving manner across distributed data sources while retaining the centralized spreading component for global recommendation generation. This direction would directly address the key concern raised in the seminar component of this thesis regarding the ethical implications of large-scale user data collection in AI-powered recommendation systems.

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Declaration

I herewith declare that this report is in full accordance with the Plagiarism Guidelines of the Faculty of Management & Technology at the GUC.

Signature: Basmala Mahrous Rasha